

BrewLedger Development Roadmap

Vision

BrewLedger bukan hanya aplikasi pencatatan stok dan transaksi.

Target akhir BrewLedger adalah menjadi platform manajemen operasional dan analitik bisnis coffee shop yang memiliki konsep serupa dengan:

- Accurate Online
- Jurnal.id
- Majoo
- Moka Backoffice

Dengan dua aplikasi client:

macOS App

Digunakan oleh:

- Owner
- Manager
- Supervisor

Fokus:

- Dashboard
- Analitik
- Monitoring bisnis
- Laporan
- Pengelolaan sistem

iOS App

Digunakan oleh:

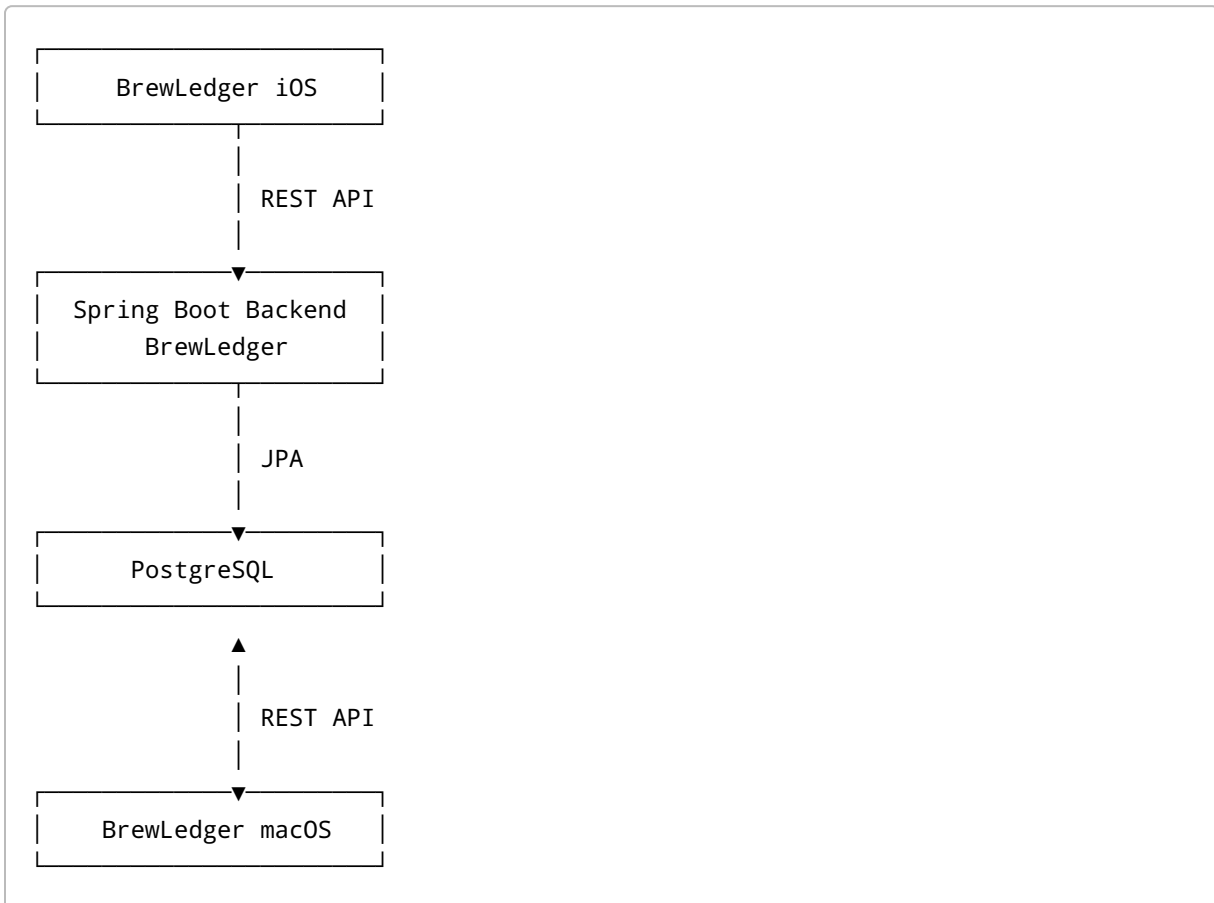
- Kasir
- Barista
- Staff Gudang

Fokus:

- Operasional harian
- Input transaksi
- Purchase order
- Stock opname

- Monitoring stok

Arsitektur Sistem



Backend Spring Boot menjadi satu-satunya sumber data.

Tidak akan dibuat backend terpisah untuk macOS maupun iOS.

Progress Saat Ini

Authentication

Selesai

Endpoint:

```
POST /api/auth/login
```

Fitur:

- JWT Authentication
- Password hashing
- Last login update
- Active user validation

Product Management

Selesai

Endpoint:

```
POST /api/products  
GET /api/products
```

Ingredient Management

Selesai

Endpoint:

```
POST /api/ingredients  
GET /api/ingredients  
GET /api/ingredients/low-stock
```

Supplier Management

Selesai

Endpoint:

```
POST /api/suppliers  
GET /api/suppliers
```

Purchase Order

Selesai

Endpoint:

```
POST /api/purchase-orders
GET /api/purchase-orders
```

Transaction

Selesai

Endpoint:

```
POST /api/transactions
```

Fitur:

- Perhitungan subtotal
- Pajak
- Total transaksi
- Pengurangan stok bahan baku

Stock Movement

Selesai

Endpoint:

```
GET /api/stock-movements
```

Fitur:

- Audit perubahan stok
 - Riwayat pembelian
 - Riwayat penjualan
-

Dashboard Summary

Versi awal selesai

Endpoint:

```
GET /api/dashboard
```

Fase Pengembangan Selanjutnya

FASE 1 — User & Role Management

User Management

Endpoint:

```
GET /api/users
GET /api/users/{id}
POST /api/users
PUT /api/users/{id}
DELETE /api/users/{id}
```

Fitur:

- Tambah user
 - Edit user
 - Hapus user
 - Reset password
-

User Activation

Endpoint:

```
PATCH /api/users/{id}/activate
PATCH /api/users/{id}/deactivate
```

Fitur:

- Aktifkan user

- Nonaktifkan user
-

Role Management

Role:

ADMIN
MANAGEMENT

Admin:

- Mengelola sistem

Management:

- Melihat analitik dan laporan
-

FASE 2 — Authorization

JWT harus menyimpan role.

Contoh:

```
{  
  "sub": "satriyadm9311",  
  "role": "MANAGEMENT"  
}
```

Implementasi:

```
@PreAuthorize("hasRole('ADMIN')")
```

```
@PreAuthorize("hasRole('MANAGEMENT')")
```

FASE 3 — Management Dashboard

Target dashboard menyerupai Accurate Online.

Activity Feed

Endpoint:

```
GET /api/dashboard/activities
```

Menampilkan:

- Login user
 - Pembuatan transaksi
 - Purchase order
 - Perubahan stok
-

Upcoming Tasks

Endpoint:

```
GET /api/dashboard/tasks
```

Menampilkan:

- PO jatuh tempo
 - Pembayaran supplier
 - Reminder stok minimum
-

Sales Trend

Endpoint:

```
GET /api/dashboard/sales-trend
```

Menampilkan:

- Grafik penjualan harian
 - Grafik mingguan
 - Grafik bulanan
-

Profit & Loss Summary

Endpoint:

```
GET /api/dashboard/profit-loss
```

Menampilkan:

- Revenue
 - HPP
 - Expense
 - Profit
-

Cash Flow

Endpoint:

```
GET /api/dashboard/cash-flow
```

Menampilkan:

- Kas masuk
 - Kas keluar
 - Saldo berjalan
-

Expense Summary

Endpoint:

```
GET /api/dashboard/expenses
```

Menampilkan:

- Biaya operasional
 - Biaya bahan baku
 - Biaya lain-lain
-

Sales Summary

Endpoint:

```
GET /api/dashboard/sales-summary
```

Menampilkan:

- Faktur lunas
 - Faktur belum lunas
 - Faktur jatuh tempo
-

Purchase Summary

Endpoint:

```
GET /api/dashboard/purchase-summary
```

Menampilkan:

- Pembelian lunas
 - Pembelian belum lunas
 - Pembelian jatuh tempo
-

FASE 4 — Business Analytics

Sales Analytics

Endpoint:

```
GET /api/analytics/sales/daily  
GET /api/analytics/sales/monthly  
GET /api/analytics/sales/yearly
```

Menampilkan:

- Penjualan per hari
- Penjualan per bulan
- Penjualan per tahun

Top Products

Endpoint:

```
GET /api/analytics/top-products
```

Menampilkan:

- Produk terlaris
- Produk kurang laku

Purchase Analytics

Endpoint:

```
GET /api/analytics/purchases
```

Menampilkan:

- Total pembelian
- Supplier ranking

Inventory Analytics

Endpoint:

```
GET /api/analytics/inventory
```

Menampilkan:

- Most used ingredient
 - Least used ingredient
 - Stock turnover
 - Low stock count
-

FASE 5 — Reporting

Sales Report

Endpoint:

```
GET /api/reports/sales
```

Filter:

- Start date
 - End date
-

Purchase Report

Endpoint:

```
GET /api/reports/purchases
```

Inventory Report

Endpoint:

```
GET /api/reports/inventory
```

Export

Format:

- PDF
 - Excel
-

FASE 6 — macOS Application

Setelah backend mencapai $\pm 95\%$.

Menu:

Dashboard
Products
Ingredients
Suppliers
Purchase Orders
Transactions
Analytics
Reports
Settings

Dashboard menggunakan widget seperti Accurate:

- Activity Feed
- Sales Trend
- Cash Flow
- Profit & Loss
- Expense Summary
- Purchase Summary
- Sales Summary

FASE 7 — iOS Application

Fokus operasional.

Menu:

Dashboard
Transactions
Purchase Orders
Stock Movement
Ingredients
Profile

Tidak menampilkan analitik kompleks.

Target Akhir

BrewLedger menjadi platform manajemen coffee shop yang memiliki:

- ✓ Inventory Management
- ✓ Purchase Management
- ✓ Supplier Management
- ✓ Sales Management
- ✓ Dashboard Monitoring
- ✓ Business Analytics
- ✓ Reporting
- ✓ Multi-role Access
- ✓ macOS Management App
- ✓ iOS Operational App